ReLoop — Thai Farmer AI

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Live demo: https://thai-farmer-agent.onrender.com

Technical Architecture · สถาปัตยกรรมระบบ

1 What technology stack powers this system?

The system is built with **FastAPI** (Python web framework), **SQLAlchemy** for database ORM, **PostgreSQL** on Render for production storage, and **httpx** for async HTTP calls. It is deployed on **Render** (Singapore region) for low latency to Thailand. The AI layer uses Anthropic's **Claude Sonnet** via the Messages API.

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2 How does the system connect to LINE?

LINE uses a **webhook model**: when a farmer sends a message, LINE's platform POSTs a JSON event to our server at `/webhook/line`. The server verifies the request is genuinely from LINE using an **HMAC-SHA256 signature** (X-Line-Signature header), then processes the message and replies using LINE's Reply API with the farmer's one-time reply token.

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3 Where is the system hosted and how reliable is it?

The system runs on **Render** (cloud platform, Singapore region) with a **managed PostgreSQL** database. The free tier has a ~50-second cold start after inactivity; the paid tier eliminates this. Render provides automatic TLS, health checks, and rollback deploys. The webhook endpoint has idempotency protection so LINE's automatic retries do not create duplicate responses.

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4 How does the system store farmer conversations?

Every inbound and outbound message is stored in PostgreSQL with full metadata: intent classification, route (grounded/refused/escalated), confidence score, model used, latency, and citation links. Farmers are stored by **pseudonymous LINE user ID only** — no phone numbers or national IDs. Sessions group messages with a 30-minute idle window.

5 Can this system scale to serve all Thai farmers?

The current architecture is stateless and horizontally scalable — Render can run multiple instances behind a load balancer. The database bottleneck can be addressed by upgrading to a paid PostgreSQL plan with connection pooling (PgBouncer). For national scale, a pgvector index would replace the in-memory cosine similarity search, and the LINE Channel would need a verified Official Account.

AI & Knowledge Base · AI และคลังความรู้

6 Which AI model is used and why Claude?

Anthropic Claude Sonnet 4.5 is used via the Messages API. Claude was chosen for its strong Thai language comprehension, instruction-following reliability, and its ability to stay grounded when given reference context. Critically, the guardrail policy is enforced in application code *independently* of the model — so Claude is a response generator, not a policy-maker.

7 What is RAG and why was it chosen over a general AI chatbot?

Retrieval-Augmented Generation (RAG) means the AI answers only from retrieved, verified documents — not from its training data alone. For agriculture, this is essential: a general AI might confidently generate plausible but wrong advice. With RAG, every answer is grounded in expert-reviewed FAQ entries with source citations, and if no relevant document exists above the confidence threshold, the system refuses rather than guesses.

8 How is the knowledge base created and maintained?

The knowledge base consists of **FaqDoc** entries with categories (rice, soil, pests, crop, government_incentive, false_news_alert). Each document has a title, Thai body text, source attribution, validity window, and status (draft/published/archived). Admins create and edit entries via the password-protected **/admin/faq** portal. New entries are immediately chunked and embedded for retrieval. *Production content must be authored by domain experts from authoritative sources such as the Department of Agriculture or Rice Department.*

9 What happens when a question has no answer in the knowledge base?

The retrieval step returns the top-K FAQ chunks scored by cosine similarity. If no chunk exceeds the minimum confidence threshold (default 0.18), the system returns a fixed Thai refusal message: *'ยังไม่มีข้อมูลที่ตรงกับคำถามนี้ในคลังความรู้'* and suggests the farmer contact a local agricultural extension officer. The AI model is never called in this case — no hallucination risk.

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10 How accurate are the AI answers?

Accuracy depends on knowledge base quality. The current demo uses a hashing embedder (character n-gram similarity) which can produce false-positive matches; the minimum score threshold mitigates this. In production, replacing the embedder with a Thai-capable model (e.g. OpenAI text-embedding-3-small) significantly improves precision. Every answer includes a source citation so admins and farmers can verify the information independently.

Guardrails & Safety · ระบบความปลอดภัย

11 How does the system prevent harmful advice about pesticide dosages?

Pesticide dosage keywords (e.g. *ปริมาณยา, อัตราการใช้สารเคมี, พาราควอต*) trigger a **hard safety gate** in the orchestrator *before* any retrieval or AI generation occurs. The system returns a fixed template refusing the specific request and directing the farmer to contact an agricultural extension officer. This gate cannot be bypassed by any prompt.

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12 What topics does the system absolutely refuse to answer?

Four categories are hard-refused: **chemical/pesticide dosage** (exact quantities, mixing instructions), **medical/human health** advice, **legal/regulatory** advice, and **financial** advice (loans, investments). These are detected by a keyword list maintained in `app/guardrails.py` covering both Thai and English terms. The list is auditable and extensible by domain experts without code changes.

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13 Can the AI hallucinate or invent agricultural facts?

No — for domain questions. The orchestrator enforces that: (1) answers must come from retrieved FAQ chunks, (2) at least one citation above the confidence threshold is required, and (3) the source is appended to every grounded answer. If those conditions cannot be met, the system refuses. For general small-talk, Claude answers conversationally with no grounding requirement, but small-talk cannot carry agricultural advice (intent classifier prevents this).

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14 Who reviews the conversations for quality?

All conversations are visible to authorised admins in the **/admin/conversations** portal. Each message is tagged with its route (grounded/refused/escalated/smalltalk) and confidence score.

Every admin view of conversation content writes an append-only **AuditEvent** record for PDPA accountability. In a production deployment, a designated expert (e.g. from the Department of Agriculture) would review flagged or low-confidence interactions weekly.

15 What happens when a high-risk question is asked?

The route is set to **'refused'** with a safe Thai template. The response never reveals partial information or hedges — it unconditionally directs the farmer to a human expert. The interaction is logged with `intent='high_risk'` so admins can monitor for patterns (e.g. a surge in pesticide-dosage questions might indicate a regional pest outbreak that warrants proactive expert communication).

LINE Integration & UX · การเชื่อมต่อ LINE

16 Why was LINE chosen as the communication platform?

LINE is **Thailand's dominant messaging app** with over 47 million active users, including extensive rural adoption. Farmers already use it daily — zero onboarding friction. The Messaging API is mature, well-documented, and supports rich menus (persistent button panels), broadcast messages, and webhook-based bot integration with strong security (HMAC-SHA256 signature verification).

17 Do farmers need to download or install anything new?

No. Farmers simply add the bot's LINE ID (**@643txfqa**) as a friend — the same way they add any contact. From that moment they can type questions in Thai and receive answers instantly. A persistent Rich Menu panel appears at the bottom of the chat with one-tap shortcuts for common topics.

18 How fast does the bot respond?

With Claude Sonnet 4.5 and a warm server, end-to-end latency (LINE → server → Claude → LINE) is typically **3–8 seconds**. The free Render plan has a cold-start delay (~50 seconds) after inactivity; the paid plan eliminates this. Database queries are indexed and the health-check endpoint keeps the service warm during active periods.

19 What languages does the system support?

The knowledge base, AI prompts, and responses are in **Thai (ภาษาไทย)**. The system prompt instructs Claude to respond in polite, accessible Thai. The intent classifier and guardrail keyword lists include both Thai and English terms, so farmers who mix languages are handled correctly. Full multilingual knowledge base support (e.g. Northern Thai dialects) is a future enhancement.

20 What happens if a farmer asks about something outside agriculture?

The intent classifier checks for domain keywords. If none are found, the message is routed to **'smalltalk'** — Claude answers conversationally (greetings, general chat) without RAG. Agriculture keywords trigger RAG retrieval. This keeps responses friendly while ensuring agricultural advice is always grounded. A farmer asking 'สวัสดีครับ' gets a warm welcome; a farmer asking about rice disease gets a verified, cited answer.

Privacy & PDPA · ความเป็นส่วนตัวและ PDPA

21 What personal data does the system store?

The system stores only the **pseudonymous LINE user ID** (a string like 'U1234abcd' assigned by LINE — not a real name, phone, or national ID). Message content, intent, route, and timestamps are stored for service quality and admin review. No biometric, location, or financial data is collected. Farmers' display names and province can optionally be stored if explicitly provided, but are not required.

22 How does the system comply with Thailand's Personal Data Protection Act (PDPA)?

PDPA compliance is built-in: (1) **Purpose-based consent** is recorded at first contact; (2) data is **pseudonymised** (LINE user ID only); (3) every admin view of conversation content writes an append-only **AuditEvent**; (4) erasure is supported with cascade deletion; (5) data is stored in Singapore (Render), which requires a DPA for cross-border transfer — this should be formalised before national deployment.

23 Can farmers request deletion of their data?

Yes. The data model supports erasure: deleting a **Farmer** record cascades to all sessions, messages, citations, and consent records. A formal erasure request workflow (e.g. farmer sending a specific message like 'ขอลบข้อมูล') can be implemented as an extension. The audit trail for admin actions is retained separately as required for accountability.

24 Is farmer conversation data shared with the government or third parties?

No by default. Conversation content is only accessible to authorised admins via the password-protected portal. If using an external LLM (Claude), message text is sent to Anthropic's API — PII should be redacted before egress in a production deployment, and a Data Processing

Agreement with Anthropic should be in place. The system is designed so the LLM provider can be swapped to an in-region option if data sovereignty requires it.

25 What is the data retention policy?

The current implementation does not enforce automatic deletion — all records persist until manually removed. For production deployment, a retention policy (e.g. delete conversations older than 2 years, retain audit events for 5 years as required by Thai accounting law) should be implemented and confirmed with legal counsel. The data architecture supports timestamped records on all tables to enable time-based purging.

Daily Briefing · ข้อมูลประจำวัน

26 What is the daily briefing and how does it work?

On each farmer's **first message of the day** (Bangkok time), the system automatically prepends a briefing to the reply. The briefing contains three parts: (1) **Chiang Mai weather forecast** (temperature, humidity, conditions via OpenWeatherMap API); (2) **government incentives** — latest subsidies and programs pulled from the knowledge base; (3) **false-news alerts** — circulating misinformation warnings curated by admins. This ensures farmers receive proactive, relevant information without needing to ask.

27 How are government incentive and false-news entries kept up to date?

Admins log in to the **/admin/faq** portal and add or archive entries under the categories `government_incentive` and `false_news_alert`. Each entry supports a **valid_from / valid_to** date window, so seasonal programs or time-limited alerts expire automatically. New entries are immediately embedded for retrieval and appear in the next day's briefing. No technical knowledge is required — the admin portal is a simple web form.

Impact & Future · ผลกระทบและอนาคต

28 How many farmers can this system realistically help?

Thailand has approximately **8.6 million farming households**. The LINE platform reaches rural areas with smartphone penetration. The current system can handle hundreds of concurrent users per instance and scales horizontally. The knowledge base currently covers rice, soil, pests, cassava, and sugarcane — expanding to Thailand's top 20 crops would address the majority of smallholder needs.

29 What is the plan to expand and improve the knowledge base?

The roadmap includes: (1) partnering with the **Department of Agriculture** and the **Rice Department** to contribute authoritative, peer-reviewed content; (2) replacing the hashing embedder with a Thai-language embedding model for better retrieval precision; (3) adding multimedia support (photos of plant disease); (4) building a structured review workflow so domain experts can approve draft entries before publication.

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30 Could this model be replicated for other provinces or crops?

Yes — the system is crop-agnostic. The knowledge base categories, guardrail keyword lists, and daily briefing sources are all configurable via the admin portal without code changes. A separate deployment for Southern Thailand (rubber, palm oil) or the Northeast (cassava, sugarcane) would require a localised knowledge base and potentially dialect-aware language handling, but the core architecture is identical.
